

Center-slice damping: the universal profile as a function of geometry

mbirjax — center-slice-noise remedy (increment 1)

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1 Setting

The reconstruction estimates a volume $x \in \mathbb{R}^{n_r \times n_c \times n_s}$ (rows $\times$ columns $\times$ slices; slices indexed $k = 0, \dots, n_s - 1$ along the rotation axis z) from a cone-beam sinogram y by minimizing the MAP cost

$$c(x) = \frac{1}{2\sigma_y^2} \|y - Ax\|_W^2 + \rho(x), \quad (1)$$

where A is the cone-beam forward projector, W the (diagonal) sinogram weights, σ_y the noise scale, and ρ the qGGMRF prior. The VCD (vectorized coordinate descent) iteration updates one *subset* of in-plane pixel positions at a time: for each pixel j in the subset it forms the cost gradient and divides by the **per-voxel curvature diagonal**

$$d_j(k) = f_h(j, k) + p_h(j, k), \quad f_h(j, k) = \frac{1}{\sigma_y^2} [A^\top W A]_{(j,k),(j,k)}, \quad (2)$$

i.e. the data term's Hessian diagonal plus the prior's local curvature, and then applies one global step size α chosen by exact line search over the subset. The resulting per-slice update array for the subset is written $\delta(\cdot, k)$.

This diagonal-plus-line-search step is accurate for updates that look like typical isolated voxels. It is systematically wrong for one particular family of updates, described next.

2 The artifact and its mechanism

The direction, and its true curvature. Fix a slice k . Let S_k denote the set of its in-support voxels, $N_k = |S_k|$, and $\mathbf{1}_k$ the indicator image of S_k . Define

$$u_k = \frac{1}{\sqrt{N_k}} \mathbf{1}_k, \quad (3)$$

the **normalized single-slice DC direction**: the image that raises every in-support voxel of slice k equally and touches nothing else ($\|u_k\|_2 = 1$). Write $f(x) = \frac{1}{2\sigma_y^2} \|y - Ax\|_W^2$ for the data term, with constant Hessian $H = \nabla^2 f = \frac{1}{\sigma_y^2} A^\top W A$ and diagonal part $D = \text{diag}(H)$, so $D_{(j,k),(j,k)} = f_h(j, k)$ from Eq. (2). (This section works with the data term alone; the prior's share of the curvature is folded in in Section 4.) The curvature of the data term along u_k is the

second derivative of the one-variable restriction $f_k(r) := f(x + r u_k)$:

$$f_k''(0) = u_k^\top H u_k = \frac{1}{\sigma_y^2 N_k} \|W^{1/2} A \mathbf{1}_k\|^2 \quad (4)$$

— computable by one weighted forward projection of a slice-indicator image.

The separable-model curvature. The step $\delta = -D^{-1}\nabla f(x)$ is not an ad-hoc scaling: it is the exact minimizer of the separable quadratic model

$$m(x + \delta) = f(x) + \nabla f(x)^\top \delta + \frac{1}{2} \delta^\top D \delta, \quad (5)$$

i.e. Newton’s method applied to a separable model of f whose Hessian is D instead of H . (“Separable model” throughout refers to this surrogate, distinct from the forward model A and the prior model ρ .) Restricting the separable model to the same line, $m_k(r) := m(x + r u_k)$ has the same slope as f_k at $r = 0$ but curvature

$$m_k''(0) = u_k^\top D u_k = \frac{1}{N_k} \sum_{j \in S_k} H_{jj} = \bar{f}_h(k), \quad (6)$$

the mean of the Hessian diagonal over slice k (this is the quantity $\bar{f}_h(k)$ of Eq. (18)). Define the **gain** as the ratio of the true to the separable-model curvature:

$$g(k) := \frac{f_k''(0)}{m_k''(0)} = \frac{u_k^\top H u_k}{u_k^\top D u_k} = 1 + \frac{u_k^\top (H - D) u_k}{u_k^\top D u_k}. \quad (7)$$

The numerator of the last term collects exactly the pairwise couplings among the slice’s voxels: $u_k^\top (H - D) u_k = \frac{1}{N_k} \sum_{i,j \in S_k, i \neq j} H_{ij}$. Along the line, the separable model’s one-dimensional minimizer sits at $r_m = -f_k'(0)/m_k''(0)$ while the true one is $r^* = -f_k'(0)/f_k''(0)$, so $r_m = g(k) r^*$. In the case that $g > 1$, the separable model’s target overshoots the true one by exactly that factor. For an axis direction e_j (an isolated voxel), $g = H_{jj}/D_{jj} = 1$ identically: the separable model is *exact* for the directions it is built from, and $g(k)$ measures how far a single-slice DC departs from this single-voxel case.

The invariant meaning of g . A raw directional second derivative such as $f_k''(0)$ carries no intrinsic meaning by itself: it scales with the normalization of the direction ($u_k \rightarrow c u_k$ multiplies it by c^2), and even at unit Euclidean norm it depends on the arbitrary per-voxel units. Concretely, rescale the voxels by any positive diagonal matrix Λ :

$$\tilde{x} := \Lambda x, \quad \tilde{f}(\tilde{x}) := f(\Lambda^{-1} \tilde{x}). \quad (8)$$

The transformed objective has Hessian $\tilde{H} = \nabla^2 \tilde{f} = \Lambda^{-1} H \Lambda^{-1}$ with diagonal part $\tilde{D} = \Lambda^{-1} D \Lambda^{-1}$, and the perturbation that raises slice k ’s in-support voxels equally is now the vector $\tilde{u}_k = \Lambda u_k$. The Euclidean-normalized curvature changes with Λ ,

$$\frac{\tilde{u}_k^\top \tilde{H} \tilde{u}_k}{\|\tilde{u}_k\|_2^2} = \frac{f_k''(0)}{\|u_k\|_2^2},$$

but the ratio g does not: its numerator and denominator are evaluated on the same vector and transform identically ($\tilde{u}_k^\top \tilde{H} \tilde{u}_k = u_k^\top H u_k$ and $\tilde{u}_k^\top \tilde{D} \tilde{u}_k = u_k^\top D u_k$), so g is a property of the objective and the perturbation alone, independent of how the voxels happen to be scaled.

Among all such rescalings there is a canonical one — the one the algorithm itself defines: $\Lambda = D^{1/2}$, so that from here on

$$\tilde{x} = D^{1/2}x, \quad \tilde{f}(\tilde{x}) = f(D^{-1/2}\tilde{x}), \quad \tilde{H} = D^{-1/2}HD^{-1/2}. \quad (9)$$

Since $\nabla \tilde{f}(\tilde{x}) = D^{-1/2} \nabla f(x)$, plain (unpreconditioned) gradient descent in $\tilde{x}$ maps back to exactly the preconditioned step in x :

$$\tilde{x}^+ = \tilde{x} - \alpha \nabla \tilde{f}(\tilde{x}) \iff x^+ = x - \alpha D^{-1} \nabla f(x), \quad (10)$$

and $\tilde{H}$ has *unit diagonal* ($\tilde{D} = I$). In these coordinates every axis direction has curvature exactly 1 (recovering the e_j remark above), and the normalized transformed direction $\tilde{u}_k / \|\tilde{u}_k\|_2 = D^{1/2}u_k / \|D^{1/2}u_k\|_2$ has ordinary Euclidean-normalized curvature

$$\frac{\tilde{u}_k^\top \tilde{H} \tilde{u}_k}{\|\tilde{u}_k\|_2^2} = \frac{u_k^\top H u_k}{u_k^\top D u_k} = g(k). \quad (11)$$

Equivalently, in metric language: the separable model (5) is the *isotropic* quadratic $\frac{1}{2} \|\delta\|_D^2$ in the inner product $\langle a, b \rangle_D = a^\top D b$ — by construction it has curvature 1 in every direction of that metric — while the objective’s actual curvature in the same metric along u_k is $u_k^\top H u_k / \|u_k\|_D^2 = g(k)$. Diagonal preconditioning is exactly the bet that the objective is nearly isotropic in its own metric; $g(k)$ is the anisotropy along u_k that survives.

Why large g breaks the iteration: the spectral statement. The iteration we analyze is the repeated application of the whitened step, Eq. (10) — the full-volume version of the update of Section 1, restricted to the data term (the subset and prior refinements are noted at the end of the paragraph and in Section 4). Because $\tilde{f}$ is quadratic, its gradient is linear in the distance from the minimizer $\tilde{x}^*$: $\nabla \tilde{f}(\tilde{x}) = \tilde{H}(\tilde{x} - \tilde{x}^*)$. Writing $\tilde{e} := \tilde{x} - \tilde{x}^*$ for the error and subtracting $\tilde{x}^*$ from both sides of Eq. (10),

$$\tilde{e}^+ = \tilde{e} - \alpha \tilde{H} \tilde{e} = (I - \alpha \tilde{H}) \tilde{e} : \quad (12)$$

each step multiplies the error by the fixed matrix $I - \alpha \tilde{H}$. $\tilde{H}$ is symmetric positive semidefinite, so it has an orthonormal eigenbasis with nonnegative eigenvalues; expanding $\tilde{e}$ in that basis, the coefficient on an eigenvector with eigenvalue λ is multiplied by $(1 - \alpha\lambda)$ at every step, so that coefficient shrinks if and only if $0 < \alpha\lambda < 2$.

Convergence is now a question about the spectrum of $\tilde{H}$: does the step size clear every eigenvalue present? The spectrum itself is out of reach — $\tilde{H}$ acts on whole volumes and is available only through projections — but it can be bounded by quantities we can compute, and the gain of Eq. (7) is exactly such a quantity. The bridge is the Rayleigh quotient $R(v) = (v^\top \tilde{H} v) / (v^\top v)$: for a symmetric matrix the extreme eigenvalues are its extreme values — in particular $\lambda_{\max}(\tilde{H}) = \max_{v \neq 0} R(v)$ — so evaluating R at *any* single test direction certifies a lower bound on $\lambda_{\max}$. Since $R(D^{1/2}u) = (u^\top H u) / (u^\top D u)$, the single-slice DC directions certify, for every slice k ,

$$\lambda_{\max}(\tilde{H}) \geq g(k). \quad (13)$$

By itself a large eigenvalue is harmless if α is small enough: shrinking α below $2/\lambda_{\max}$ would make every coefficient contract. But the algorithm does not choose α that way. The update

direction is $\tilde{v} := -\nabla \tilde{f}(\tilde{x})$, and the line search minimizes the quadratic $\alpha \mapsto \tilde{f}(\tilde{x} + \alpha \tilde{v})$ exactly; setting its derivative to zero gives

$$\alpha = \frac{\tilde{v}^\top \tilde{v}}{\tilde{v}^\top \tilde{H} \tilde{v}} = \frac{1}{R(\tilde{v})} \quad (14)$$

(the same identity holds per subset for the subset-restricted quantities). The step size is therefore matched to the curvature of the direction the iteration is currently moving along — not to the largest curvature present. Combining Eqs. (13) and (14): whenever some slice satisfies

$$g(k) \geq 2 R(\tilde{v}), \quad (15)$$

we have $\alpha \lambda_{\max} \geq \alpha g(k) \geq 2$, so $I - \alpha \tilde{H}$ has an eigendirection whose coefficient satisfies $|1 - \alpha \lambda| \geq 1$: some component of the error fails to shrink, and grows when the inequality is strict. In words: the error stops contracting whenever the gain along some slice’s DC direction is at least twice the Rayleigh quotient of the direction the line search is serving. The dangerous regime is the ordinary one — when the residual driving the step is spread across typical directions, $R(\tilde{v})$ is near 1 (the regime the separable model was built for), and Eq. (15) is met by any slice with $g(k) \gtrsim 2$.

Two questions address the relevance of this analysis, with answers that are primarily empirical and supported by well-defined measurements taken up in Section 3:

1. How large is $g(k)$ for the slices near the source plane?
2. Is the bound (13) nearly attained by u_k ? I.e., is the single-slice DC family a strongly non-contracting eigendirection?

From non-contraction to the visible artifact. Two different objects must be kept apart here. The first is the minimizer $\hat{x}$ of the cost: it is defined by the cost alone, so nothing we do to the *step* can move it — Section 5 shows that any positive per-direction rescaling of the update leaves the fixed point of the iteration exactly at $\hat{x}$. The second is the iterate x_t the algorithm actually produces after t steps, and it is here that non-contraction does its damage: the artifact is a property of the *path*, not of the minimizer.

Consider the coefficient of the error $e_t = x_t - \hat{x}$ along the non-contracting eigendirection. In the idealized analysis above — the full-volume operator and a fixed α — that coefficient would grow without bound. The real iteration departs from the idealization in three ways, and together they convert unbounded growth into bounded non-convergence. First, α is re-chosen at every subset and fluctuates with the composition of the residual, so the per-step multiplier $(1 - \alpha \lambda)$ varies around magnitude one rather than sitting at a fixed amplifying value. Second, each step applies the operator of one randomly assigned *subset*, not the full volume, which further randomizes the multiplier. Third, each subset step differs from the full-gradient step, and that difference — driven by the uneven split of the residual, measurement noise included, across subsets — acts as a fresh perturbation with a component along the eigendirection. The coefficient therefore evolves as a stochastic recursion with per-step gain near one and persistent excitation: it neither decays nor explodes, but wanders in a band whose width is set by the excitation.

At any stopping iteration the error along the eigendirection thus has two parts: whatever the initialization placed there, which a contracting iteration would have removed and this one preserves; and the fluctuating part continually re-injected by the subset iteration. Granting the

identification in question 2 above — the eigendirection concentrated on the single-slice DC components of a band of slices near the source plane — this error appears as elevated, slowly clearing noise in exactly those slices: the “plume.” The minimizer $\hat{x}$ contains no plume; the plume lives entirely in the path. That is why the remedy of Section 4 modifies the step and not the cost function (1).

Where geometry enters: why g peaks at the source plane. Everything so far defines $g(k)$ and explains what a large value does; nothing yet says *which* slices should have one. That is where the scanner’s geometry enters. By Eq. (7), $g(k)$ exceeds 1 by the total pairwise coupling among the slice’s voxels, measured against the diagonal, and each coupling is explicit:

$$H_{ij} = \frac{1}{\sigma_y^2} \sum_m w_m A_{mi} A_{mj} \quad (16)$$

— a weighted inner product, over all detector measurements m in all views, of the two voxels’ projection footprints. H_{ij} is large when voxels i and j project onto the *same detector cells* in *many views*. The question “which slices have large gain” therefore reduces to a geometric one: for which slices do pairs of same-slice voxels keep landing on the same detector cells, view after view?

A detector cell is the intersection of a channel (the transaxial coordinate) and a row (the axial coordinate), so a pair contributes to Eq. (16) only in views where it is aligned in both. Channel alignment is the indiscriminating part: two voxels of the same slice are channel-aligned in the few views whose rays pass through both, and this holds for every pair in every slice — it cannot single out the source plane. The discrimination comes entirely from the rows. A voxel at axial distance z from the source plane projects to detector row $\approx z m / \Delta_{\text{row}}$, where the magnification m (the ratio of the source–detector distance to the source–voxel distance) varies with the voxel’s in-plane position and with the view. For a source-plane voxel, $z = 0$: its row is the *source row* — the detector row at the height of the source — regardless of its position, its magnification, or the view. So when a source-plane pair reaches its channel-aligned views, the row condition holds automatically: in every such view, for every pair in the slice, all the way around the orbit. Nothing cancels, and the pairwise sum in Eq. (7) accumulates coherently.

For a slice at $|z| > 0$ the row condition starts to fail. When two of its voxels are ray-aligned, they sit at different distances from the source — hence at different magnifications — and their rows differ by $\approx z \Delta m / \Delta_{\text{row}}$, a separation that grows with the slice’s distance from the plane and with the pair’s in-plane separation. Once it exceeds a detector row, the pair’s overlap is gone. As $|z|$ grows, an ever-smaller neighborhood of pairs keeps both alignments, and the pairwise mass in the numerator of Eq. (7) drains away.

The geometric prediction is therefore qualitative but sharp: $g(k)$ peaks at the source-plane slice k_0 , decays with $|k - k_0|$ at a rate set by the cone’s divergence (a wider cone separates the rows of ray-aligned pairs faster, isolating the source plane more sharply from the bulk), and approaches 1 far away. The actual profile — its height, width, and decay law — is a measurement, taken up in Section 3. One structural remark completes the picture: the argument used only the row and channel alignment of voxel pairs within a slice, so it applies verbatim to any *subset* of the slice’s voxels — the pairs that remain still share rows at $z = 0$. Every VCD subset therefore sees the same coherence through its own in-plane mean, which is why the overshoot acts at every subset granularity and why the remedy of Section 5 targets each subset’s mean component.

3 Quantities

symbol	meaning	source
n_s	number of recon slices	recon shape
Δz	slice pitch = <code>delta_voxel</code> $\times$ <code>voxel_slice_aspect</code>	geometry
z_c	the source’s axial coordinate (source and detector are fixed; the object rotates); the source plane $z = z_c$ meets the detector in a single row, the source row	geometry
<code>recon_slice_offset</code>	z of the volume center relative to $z = 0$	geometry
k_0	source-plane position in slice-index units, Eq. (17)	derived
Δk	$ k - k_0 $, slice distance from the source plane	derived
$g(\Delta k)$	the per-slice gain: measured curvature-gain of the slice-mean mode, Table 1	measured
$\bar{f}_h(k)$	per-slice mean data curvature, Eq. (18)	this recon
$\bar{p}_h(k)$	per-slice mean prior curvature, Eq. (19)	this recon
$s(k)$	the damping profile, Eq. (20)	derived
$\delta(\cdot, k)$	a subset’s update array at slice k	iteration
$\bar{\delta}(k)$	in-plane mean of $\delta(\cdot, k)$ over the subset’s pixels	iteration
$s_{\min}$	damping-depth floor, 0.1	measured
θ_{off}	disarm threshold on the drive ratio, 1.3	measured

The source-plane slice position. With slice k centered at $z(k) = (k - \frac{n_s-1}{2})\Delta z + \text{recon_slice_offset}$, solving $z(k_0) = z_c$ gives

$$k_0 = \frac{n_s - 1}{2} - \frac{\text{recon_slice_offset} - z_c}{\Delta z}. \quad (17)$$

The grid snap (`snap_z0_to_slice_center`, default on) adjusts `recon_slice_offset` by at most half a slice so that k_0 is an integer — the mode then concentrates in one slice, where both the damping and the trigger read it most cleanly. On an unsnapped grid that leaves k_0 between slices, the two straddling slices both carry the mode and both receive the cap treatment.

The per-slice gain. $g(\Delta k)$ is the measured response of the diagonally-preconditioned data operator to a single-slice perturbation, expressed relative to a far-from-center slice, tabulated against Δk and interpolated log-log (approximately $g \sim \Delta k^{-0.7}$):

Δk	0	1	2	3	5	8	12	20	30	60	100	200
g	71.1	62.8	48.7	38.4	25.1	16.6	11.4	6.90	4.69	2.36	1.42	1.0

Table 1: The measured gain table (single-slice probes, widest-cone dataset). The *shape* is near-universal across measured datasets; only the peak amplitude grows with cone angle.

The per-reconstruction curvature means. These are computed once at the start of `recon()` from this reconstruction’s own operators and initial state:

$$\bar{f}_h(k) = \frac{1}{\sigma_y^2} \cdot \text{mean}_{j: f_h(j,k) > 0} [A^\top W A]_{(j,k),(j,k)} \quad (\text{in-support pixels only}), \quad (18)$$

$$\bar{p}_h(k) = \text{mean}_{j \in \mathcal{S}} \left. \frac{\partial^2 \rho}{\partial x_{j,k}^2} \right|_{x=x^{(0)}} \quad (\mathcal{S} \text{ a fixed random subsample of in-support pixels}), \quad (19)$$

where $x^{(0)}$ is the initial recon (typically the FDK direct reconstruction). A few thousand sampled pixels estimate the slice mean of the prior curvature well, so the cost of Eq. (19) is negligible.

4 The damping profile

The diagonal preconditioner understates the slice-mean mode’s curvature by the factor g *on the data term only*: the prior is local, so its share of the curvature is correctly represented by the diagonal. The effective curvature of the mode at slice k is therefore $g(\Delta k) \bar{f}_h(k) + \bar{p}_h(k)$, while the step was computed as if it were $\bar{f}_h(k) + \bar{p}_h(k)$. Multiplying the mode’s step by the ratio of the two restores the correct step size for that mode, relative to an ordinary bulk slice:

$$s(k) = \text{clip}_{[0,1]} \left(\frac{\bar{f}_h(k) + \bar{p}_h(k)}{g(|k - k_0|) \bar{f}_h(k) + \bar{p}_h(k)} \right), \quad s(k) \leftarrow \max(s(k), s_{\min}) \text{ wherever } s(k) < 1. \quad (20)$$

The limits behave correctly without special cases:

- **Bulk slices** ($\Delta k \gtrsim 200$): $g = 1 \Rightarrow s = 1$ — no damping.
- **Data-starved end slices**: $\bar{f}_h \rightarrow 0 \Rightarrow s \rightarrow 1$ — the prior already governs the step there, and the prior’s curvature is not understated.
- **The cap on a wide-cone dataset**: $\bar{f}_h \gg \bar{p}_h \Rightarrow s \approx 1/g \approx 0.014$, which is caught by the floor $s_{\min} = 0.1$. The floor is the measured point of diminishing returns: damping deeper than 0.1 saturated the benefit and only added transient cost.

Cone-angle dependence is deliberately *implicit*: the table’s shape is the near-universal part, and the amplitude differences across datasets are partly absorbed by each reconstruction’s own $\bar{f}_h/\bar{p}_h$ balance, with the trigger of Section 6 standing the damping down entirely where the mode is not driven.

5 Rank-one application, and why it is safe

The profile is *not* applied to the whole update. Only the coherent in-plane-mean component of each slice’s update — the runaway subspace — is rescaled:

$$\tilde{\delta}(\cdot, k) = \delta(\cdot, k) + (s(k) - 1) \bar{\delta}(k), \quad \bar{\delta}(k) = \text{mean}_{j \in \text{subset}} \delta(j, k). \quad (21)$$

Equivalently $\tilde{\delta} = (I + (s(k) - 1) P_{\text{mean}}) \delta$ per slice, with P_{mean} the projector onto the in-plane constant vector. Two facts justify this form:

It targets exactly the right subspace. The top eigenmode of the preconditioned normal operator, computed by power method, *is* the coherent slice-profile mode: at its peak slice the in-plane mean carries essentially all of the mode energy. Scaling only that component damps the runaway mode while per-voxel structure continues to move at full speed — measured, this converges structure near the cap $\sim 10\times$ closer to the converged reference at 15 iterations than the undamped baseline, while removing the noise plume.

It cannot move the answer. For $s(k) > 0$ the operator $I + (s - 1)P_{\text{mean}}$ is positive definite, so Eq. (21) is a positive-definite rescaling of the preconditioner. Such a rescaling changes the

path of the iteration but not its stationary points: the converged solution of the damped and undamped iterations is the same minimizer of Eq. (1). This is the key safety property — the remedy shapes convergence dynamics and cannot bias the reconstruction.

6 The decide-once trigger

The knob is armed by default, so it must stand down automatically on data whose mode is never driven. During the first two iterations the updater also accumulates the *raw* (pre-damping) per-slice drive

$$r_i(k) = \left(\sum_{\text{subsets in iteration } i} \|\delta_{\text{raw}}(\cdot, k)\|_2^2 \right)^{1/2}, \quad i = 0, 1, \quad (22)$$

and at the end of each iteration forms the cap-to-bulk ratio

$$R_i = \frac{\max_{k \in \text{cap}} r_i(k)}{\text{median}_{k \in \text{bulk}} r_i(k)}, \quad (23)$$

where the cap is the slice (or straddling pair) at k_0 and the bulk band is $\Delta k \in [80, 150]$ away from the volume ends. After the second reading one latched decision is made: if $\max(R_0, R_1) < \theta_{\text{off}} = 1.3$ the damping is **disarmed** for the rest of the run. The statistic must be the raw (pre-damping) drive — the damped, committed updates read ~ 0.2 – 0.3 at the cap and would release instantly. Measured bands: immune datasets ≤ 0.93 , plume datasets ≥ 1.53 on this statistic, so 1.3 sits in the gap with margin on both sides; two armed iterations on an immune dataset are measured indistinguishable from never damping. The decision is made once and never revisited (a symmetric on/off threshold was measured to chatter), and continued runs reuse it.

Scope and provenance. The formula above is the **universal path**, used when the volume can carry the bulk reference band (roughly $n_s \gtrsim 180$ around the source plane). Thin row-cropped slabs instead use the adaptive *seed-then-convert* path (increment 2), which measures its own drive shape in place of Table 1. The measured basis and the full validation record are in `plans/bugs_and_artifacts/center slice noise/final_findings.md`; the developer overview is `docs/source/dev_center_slice_noise.rst`.